Advanced Telecommunication Technologies and Management

MASTERS 2

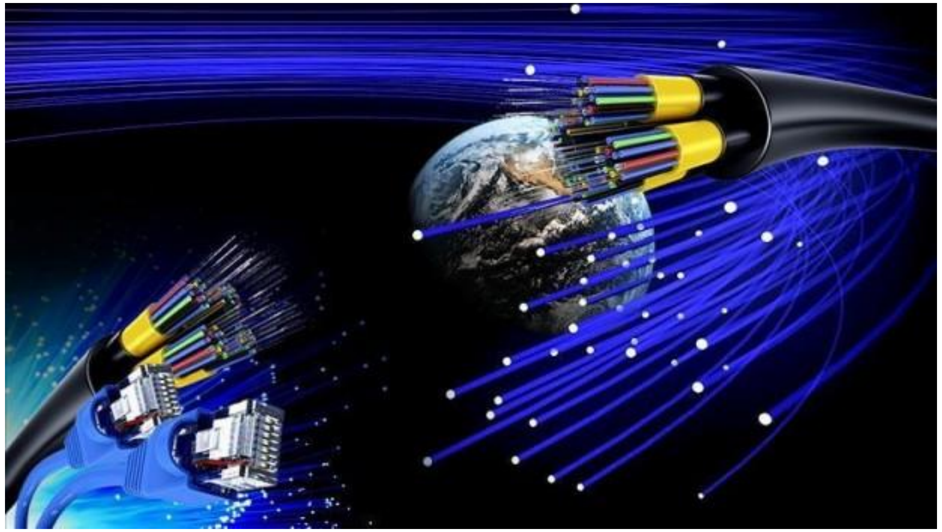
FACULTY OF ECONOMIC SCIENCES AND BUSINESS ADMINISTRATION
BUSINESS INFORMATICS DEPARTMENT
LEBANESE UNIVERSITY
Dr. Khouloud AL-SAMROUTH

Course objectives

- Computer Communication Networks
- Signal Theory
- Digital Communications
- Telecommunication Technologies
- Minor Project (Telecom. Technology & Management)

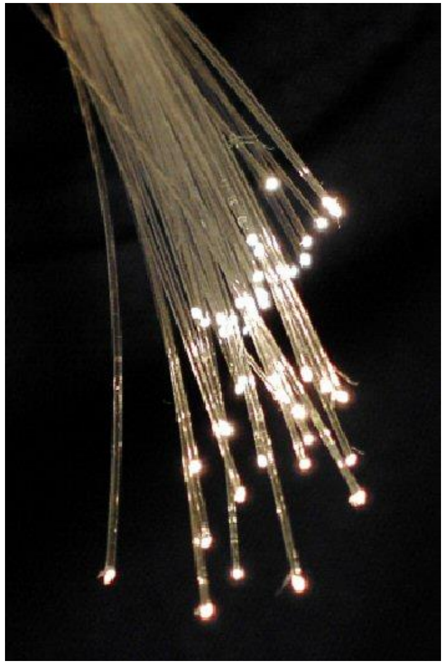
Outline

- Introduction
- What are Optical Fibers?
- Structure of fiber optic
- Working principle of fiber optic
- Optical fiber communication system
- Advantages / Disadvantages of Optical fiber
- Applications of Optical fiber
- Conclusion



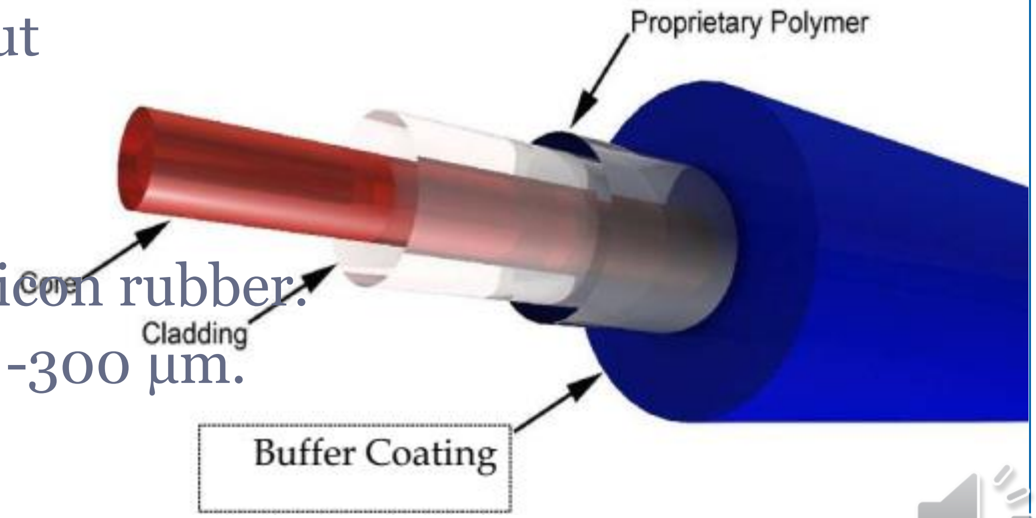
What is Optical Fiber?

- An optical fiber is a hair thin cylindrical fiber of glass or any transparent dielectric medium.
- Wave guides used for optical communication over long distances.
- Speed: Current record = 15.5 tbps



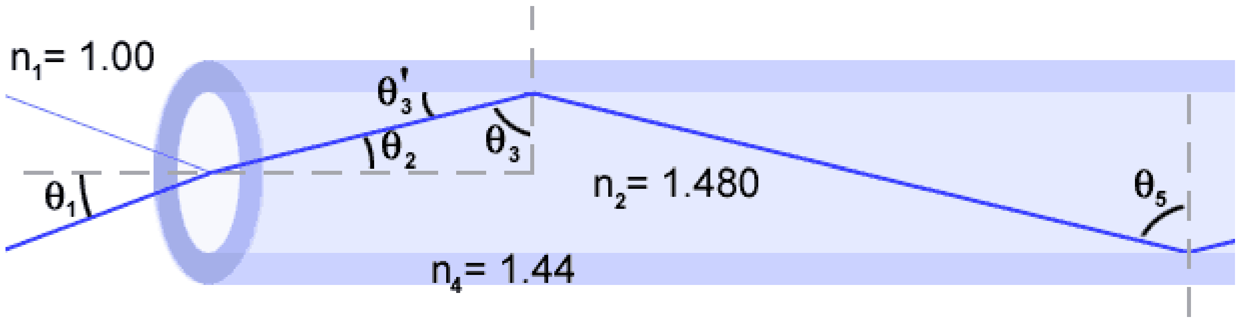
Structure of Fiber Optic

- Core
  - Central tube of very thin size (core diameter varies from 5 μm to 100 μm)
  - Made up of optically transparent dielectric medium (i.e: reflecting index n = 1.43)
  - Carries the light form transmitter to receiver.
- Cladding
  - Outer optical material surrounding the core having reflecting index lower than core.
  - Helps to keep the light within the core throughout the phenomena of total internal reflection.
- Buffer Coating
  - Plastic coating that protects the fiber made of silicon rubber.
  - The typical diameter of fiber after coating is 250 -300 μm.



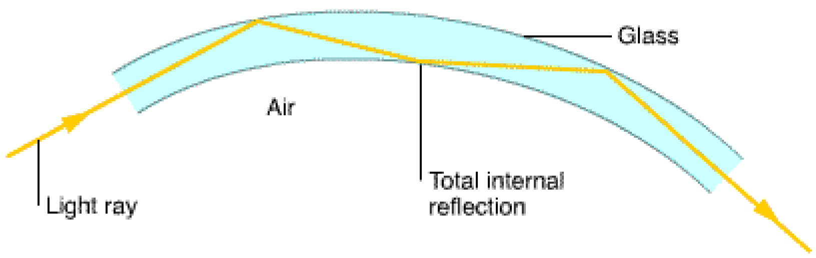
Working Principle

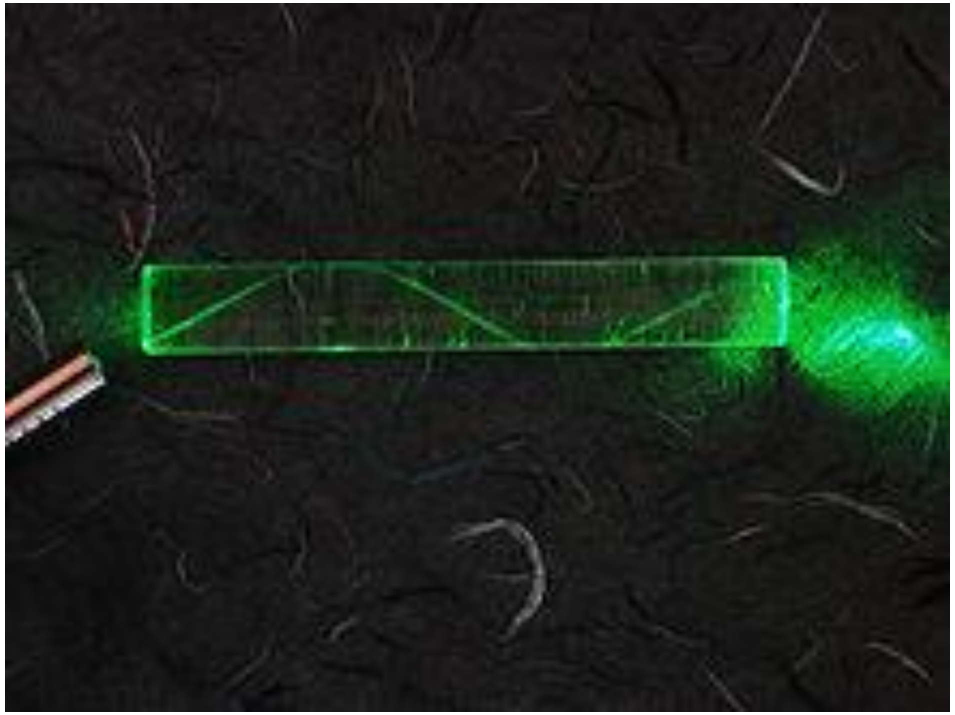
- Total Internal Reflection
  - When a ray of light travels from a denser to a rarer medium such that the angle of incidence is greater than the critical angle, the ray reflects back into the same medium this phenomena is called total internal reflection.
  - In the optical fiber the rays undergo repeated total number of reflections until it emerges out of the other end of the fiber, even if the fiber is bent.



Working Principle

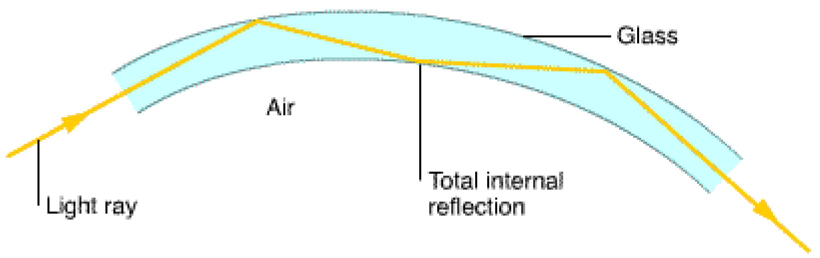
- Total Internal Reflection in a Fiber Optic

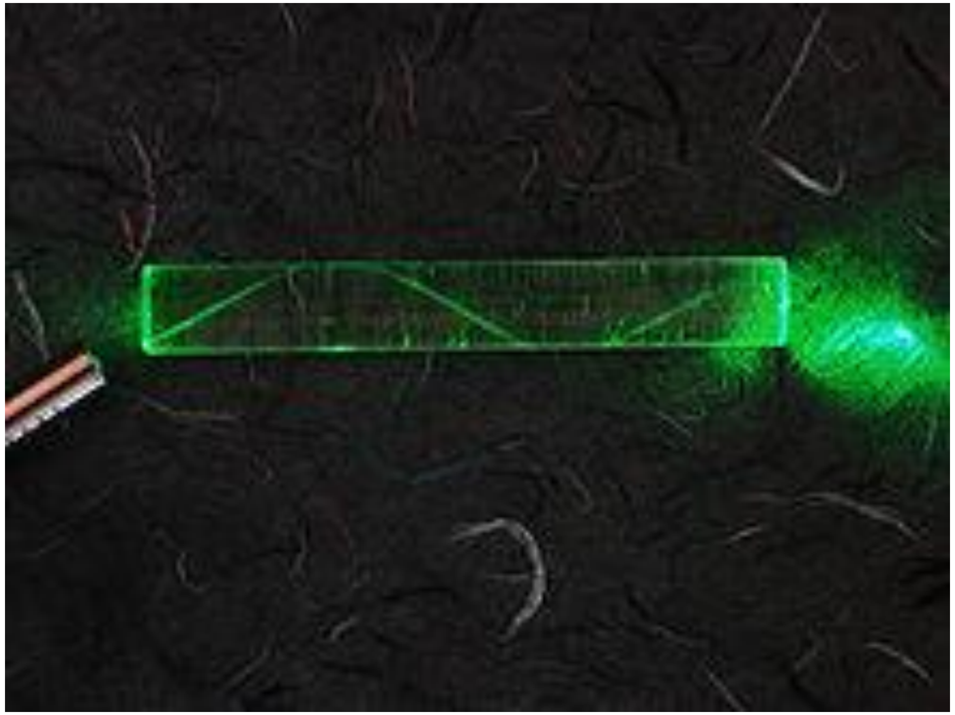




Working Principle

- Total Internal Reflection in a Fiber Optic





Working Principle

- Numerical Aperture of Optical Fiber
  - The significance of NA is that light entering in the cone of semi vertical angle i_m only propagates through the fiber.
  - The higher the value of i_m or NA more is the light collected for propagation in the fiber.
  - Numerical aperture is thus considered as a light gathering capacity of an optical fiber.

$$NA = \sqrt{n_1^2 - n_2^2}$$

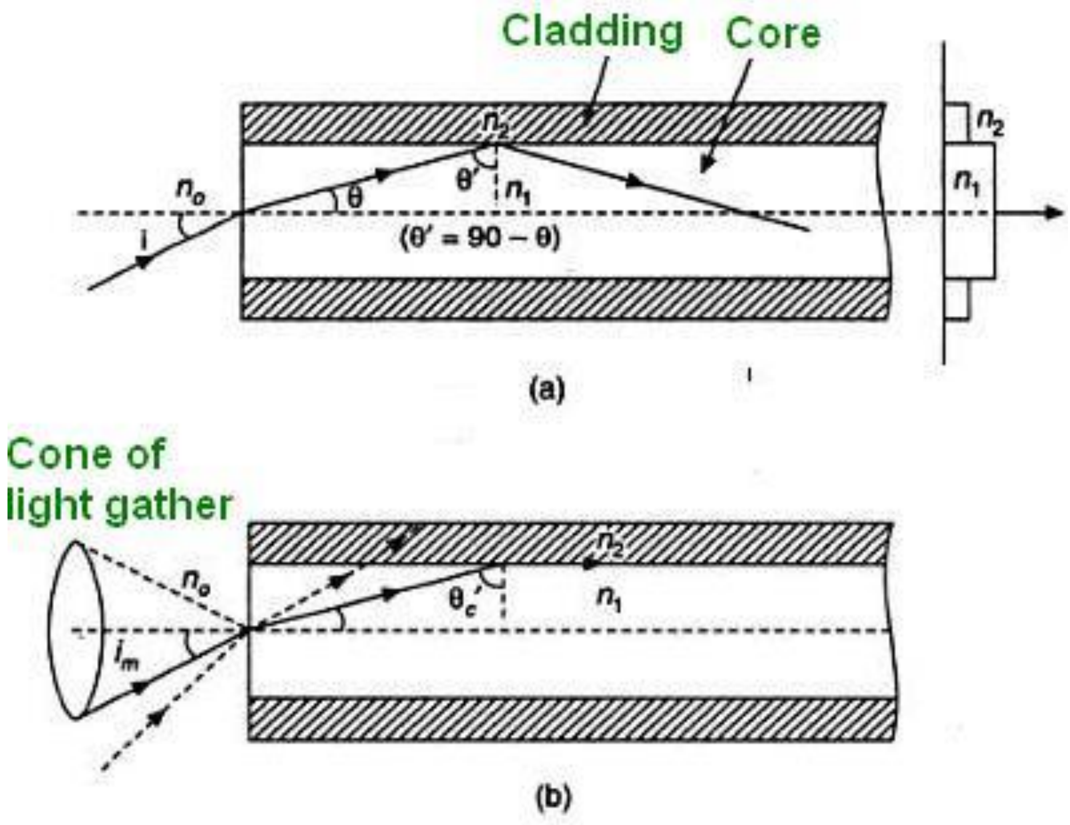
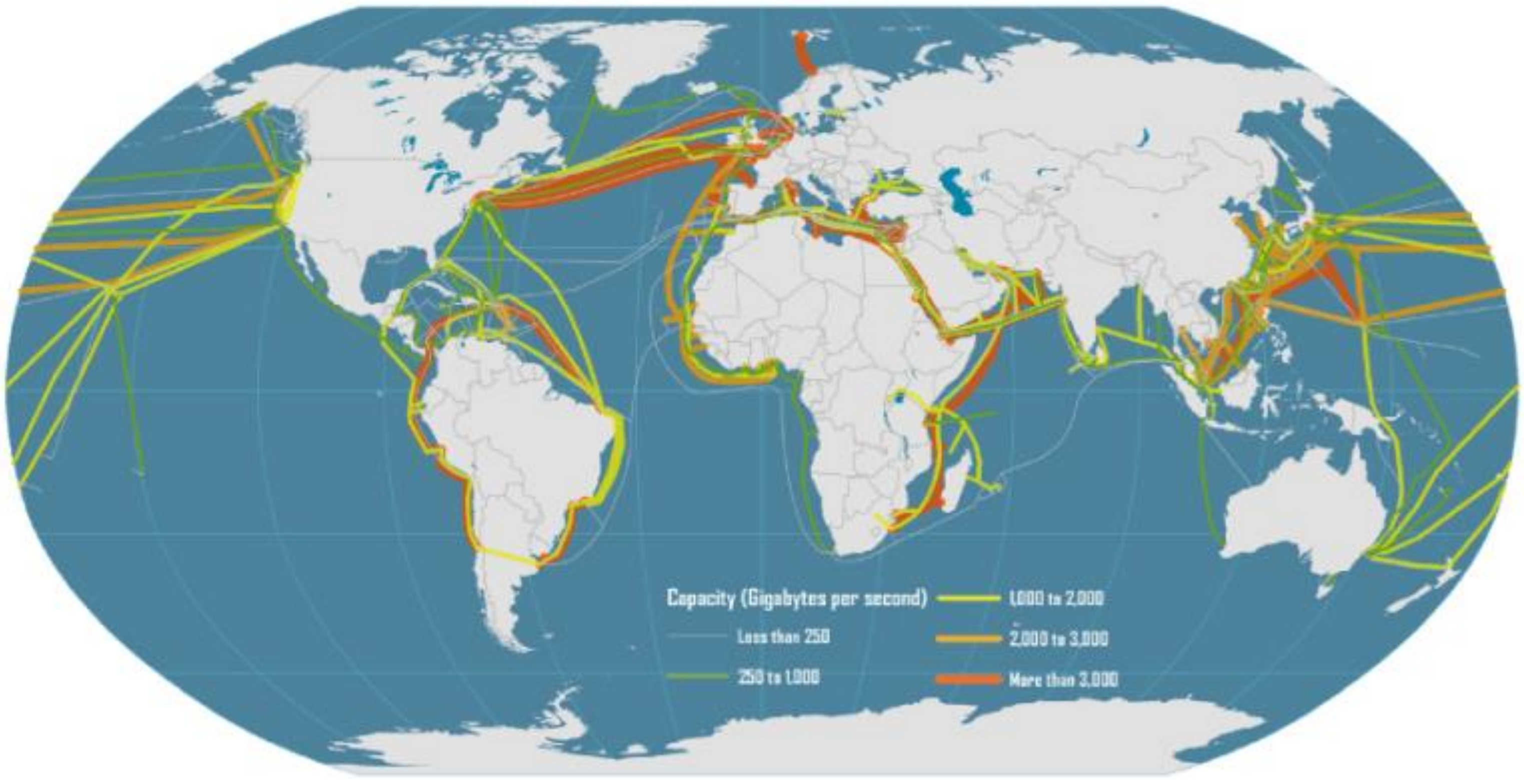
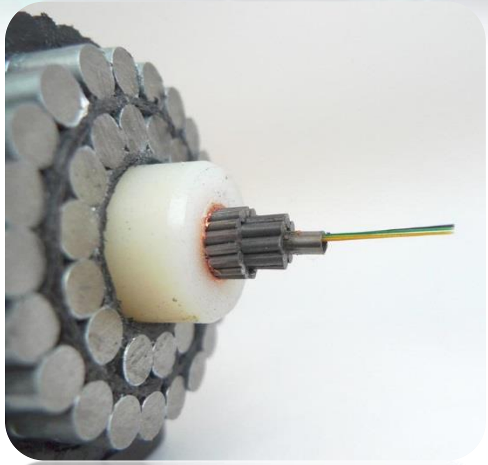


Figure 3.

Working Principle

- tat-8 transatlantic cable



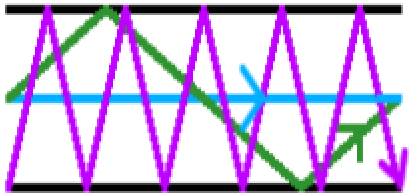


Classification of Optical Fiber

- Optical fiber is classified into two categories based on :
  - The number of modes
    - Single mode fiber (SMF)
    - Multi-mode fiber (MMF)
  - The refractive index
    - Step-index optical fiber
    - Graded-index optical fiber

Single mode fiber

Multi mode fiber



Classification of Optical Fiber

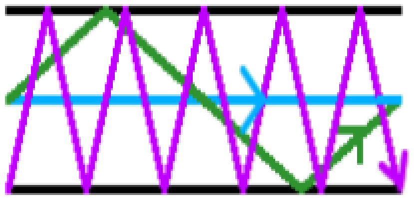
- Single mode fibers
  - Only one mode can propagate through the fiber.
  - Small core diameter (5µm) and high cladding diameter (70µm)
  - Very small difference between the refractive index of core and cladding.
  - No dispersion i.e. no degradation of signal during travelling through the fiber.
  - The light is passed through the single mode fiber through laser diode.
  - 100 km : cable TV, internet and telephone cables

Single mode fiber

Classification of Optical Fiber

- Multi mode fiber
  - Allows a large number of modes for the light ray travelling through it.
  - The core diameter is (40μm) and that of cladding is(70μm)
    - 10 times bigger than one in a single-mode cable
  - The relative refractive index difference is larger than single mode fiber.
  - Signal degradation due to multimode dispersion.
  - Not suitable for long distance communication due to large dispersion and attenuation of the signal.
    - to link computer networks together<3km, LAN

Multi mode fiber



Classification of Optical Fiber

- Multi mode fiber

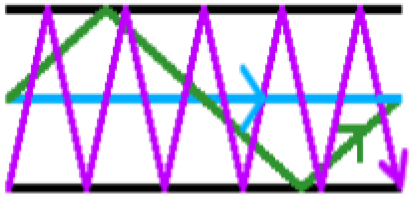
$$M_n = \frac{V^2}{2}$$

$$V = \left[\frac{2\pi a}{\lambda}\right] * NA$$

  - Where

    M_n : number of modes
    V : normalized frequency
    a : fiber core radius
    λ : operating wavelength
    NA : numerical aperture

Multi mode fiber



- Exercise
  - What is the maximum core radius for a fiber if it is to operate in single mode at a wavelength of 1550 nm if the *NA* is 0.12 and *V* = 2.405?

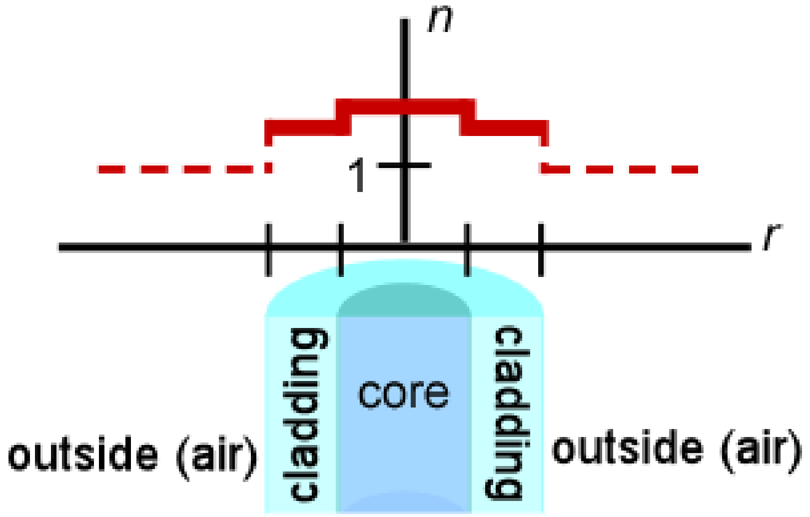
$$V = \left[\frac{2\pi a}{\lambda}\right] * NA$$

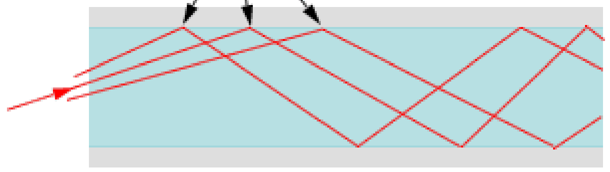
$$2.405 = \left[\frac{2 * 3.14 * a}{1550 * 10^{-9}}\right] * 0.12$$

$$a = 4.95 * 10^{-6} = 4.95\mu m$$

Classification of Optical Fiber

- Step index fiber
  - The refractive index of core is constant
  - The refractive index of cladding is also constant
  - The light rays propagate through it in the form of meridional rays which cross the fiber axis during every reflection at the core cladding boundary

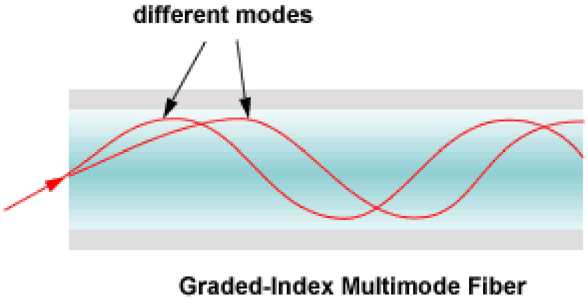


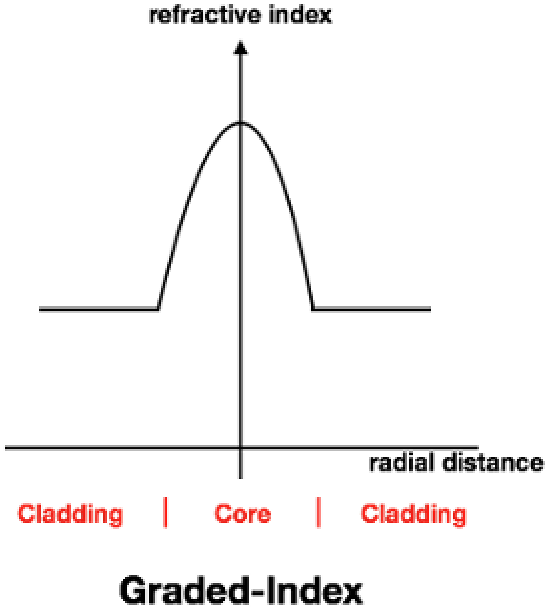


Step-Index Multimode Fiber

Classification of Optical Fiber

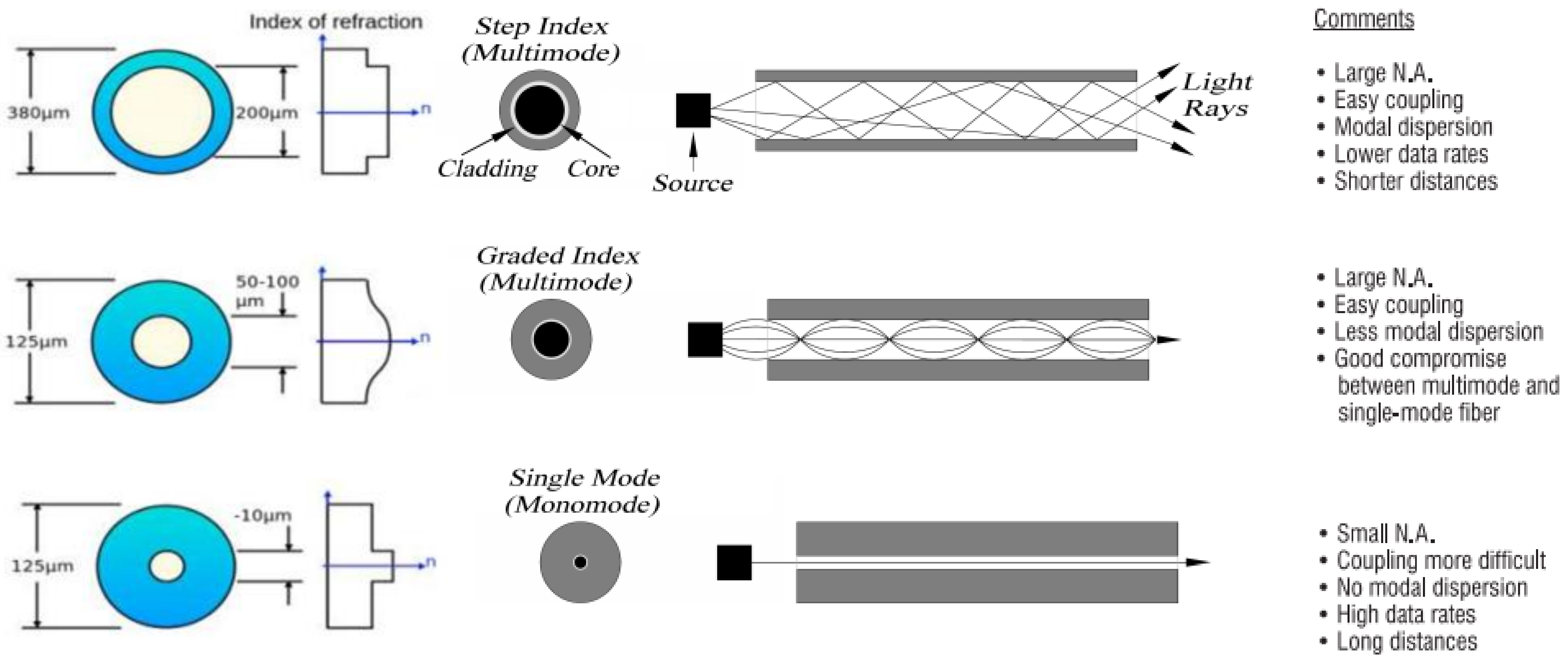
- Graded Index fiber
  - Fiber core has a non uniform refractive index that gradually decrease from the center towards the core cladding interface.
  - The cladding has a uniform refractive index
  - The light rays propagate through it in the form of skew rays or helical rays.





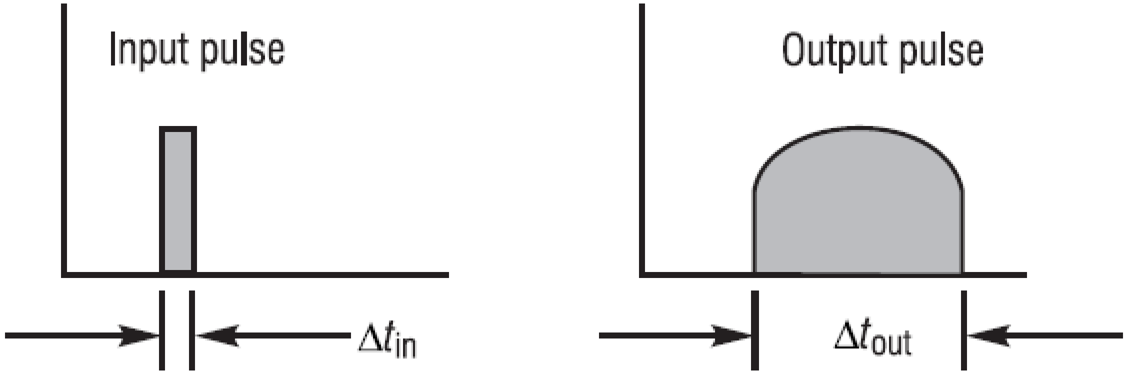
Classification of Optical Fiber

- Difference between different Fiber Optic classes



Dispersion

- Dispersion (Δt) is defined as pulse spreading in an optical fiber.
- As a pulse of light propagates through a fiber, elements such as numerical aperture, core diameter, refractive index profile, wavelength, and laser line width cause the pulse to broaden.
- This poses a limitation on the overall bandwidth of the fiber



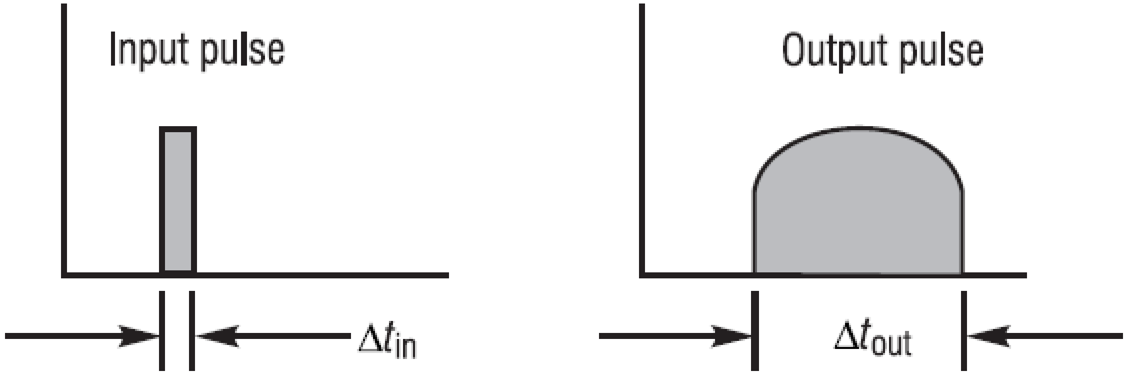
Pulse broadening caused by dispersion

Dispersion

- Dispersion

$$\Delta t \;=\; (\Delta tout - \Delta tin)^{1/2}$$

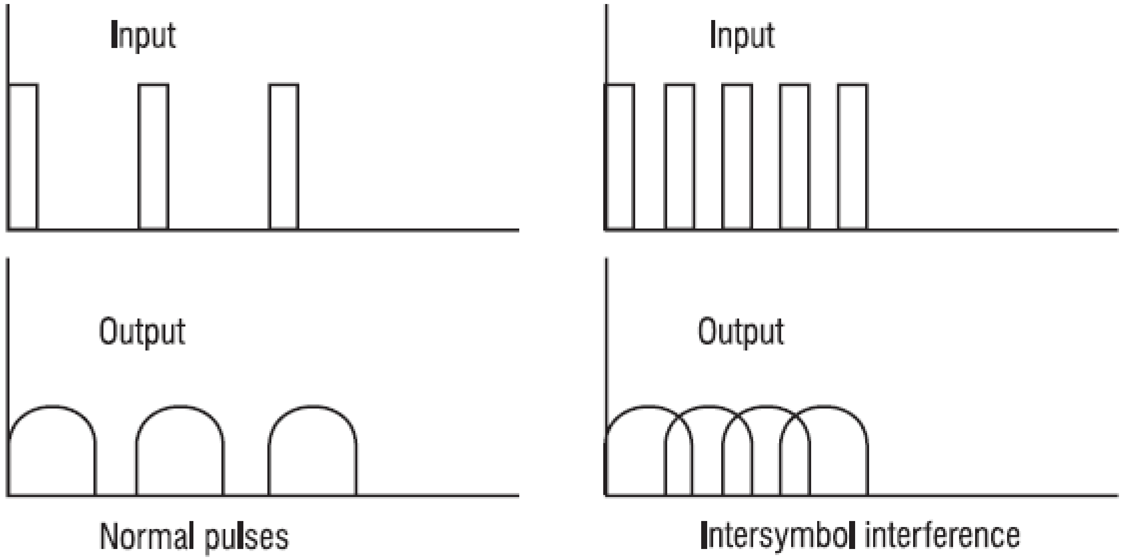
$$\Delta t_{total} = L \times (Dispersion/km)$$



Pulse broadening caused by dispersion

Dispersion

- The overall effect of dispersion on the performance of a fiber optic system is known as intersymbol.
- Intersymbol interference occurs when the pulse spreading caused by dispersion causes the output pulses of a system to overlap, rendering them undetectable.



Intersymbol interference

Dispersion

- Modal dispersion (ps/km)
  - Pulse spreading caused by the time delay between lower-order (modes or rays propagating straight through the fiber close to the optical axis) and higher-order modes (modes propagating at steeper angles).
- Chromatic dispersion $(ps/km \bullet nm)$
  - Pulse spreading due to the fact that different wavelengths of light propagate at slightly different velocities through the fiber.
    - due to the wavelength dependency on the index of refraction of glass.
    - due to the physical structure of the waveguide.

Dispersion

- Dispersion

$$\Delta_{tot} = [(\Delta t_1)^2 + (\Delta t_2)^2 + \ldots\ldots + (\Delta t_n)^2\ ]^{1/2}$$

- where Δt_n represents the dispersion due to the various components that make up the system.

Dispersion

- The approximate bandwidth of a fiber can be related to the total dispersion by the following relationship

$$BW = 0.35/\Delta t_{total}$$

- The transmission capacity of fiber is typically expressed in terms of *bandwidth × distance.*
  - for a typical 62.5/125-µm (core/cladding diameter) multimode fiber operating at 1310 nm might be expressed as 600 MHz • km.

Dispersion

- Exercise
  - A 2-km-length multimode fiber has a modal dispersion of 1 ns/km and a chromatic dispersion of 100 ps/km • nm. If it is used with an LED of linewidth 40 nm.

a) What is the total dispersion?

b) Calculate the bandwidth (BW) of the fiber.

$$\Delta t_{modal} = 2\,km \times 1\,ns/km = 2\,ns$$

$$\Delta t_{chromatic} = (2\,km) \times (100\,ps/km \bullet nm) \times (40\,nm) = 8000\,ps = 8\,ns$$

$$\Delta t_{total} = (\,(2\,ns)\verb|^|2 + (8\,ns)2)\verb|^|1/2 = 8.24\,ns$$

$$BW = 0.35/\Delta t_{total} = 0.35/8.24\,ns = 42.48\,MHz$$

$$Expressed\ in\ terms\ of\ (BW \bullet km), we\ get$$

$$(BW \bullet km) = (42.5\,MHz)(\,2\,km) \sim\!-85\,MHz \bullet km.$$

THANK YOU!